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Sampled or Stated? Post-Training Moves a Language Model's Differential Diagnosis from Its Samples to Its List

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Abstract

A language model's differential diagnosis can be read from the ranked list it states in one answer or from the spread of its answers over repeated samples. Pass@k, self-consistency and sampling-based uncertainty use the second. We compare the two readouts in base/post-trained pairs from two open families at 7–32B (Qwen2.5-7B and -32B, Qwen3-8B, and every released stage of OLMo 3 7B and OLMo 3.1 32B) and in three medical models fine-tuned from one parent, with 32 samples per case on DDXPlus, whose patients come with a reference differential, and on rare-disease cases from Phenopacket Store. Post-training consistently moves the useful readout from the samples to the list. The sampled differential collapses: post-trained models trail their base models by 4–28 points at pass@32 and give 1.3–2.6 distinct top answers per 32 samples against 3.8–8.3; majority voting stops helping them, and agreement between samples becomes a weaker error signal. Meanwhile the stated list improves: in every family and size it moves closer to the reference differential while the samples move away (for OLMo 3 7B when asked for a full distribution), and post-trained Qwen models name can't-miss conditions 9–13 points more often. Which readout is the better error signal depends on the model, but a cheap audit predicts it: across 56 models, 100 cases with five samples each pick the better one in 88% of draws. Which post-training stage narrows the samples most depends on the recipe. Claims about a model's diagnostic breadth should say which readout they use.

Data and Code Availability. All data are public: DDXPlus (Fansi Tchango et al., 2022) (synthetic patients) and Phenopacket Store v0.1.27 (Danis et al., 2025) (cases from published literature). All models are public checkpoints (Table 4). Code, prompts and sampled outputs will be released with the camera-ready version.

Institutional Review Board (IRB). This work uses only public synthetic or previously published, de-identified data and involves no human subjects, so it did not require IRB review.

1. Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. A language model's differential can be read in two ways: from the list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting (Wang et al., 2023) and sampling-based uncertainty such as semantic entropy (Farquhar et al., 2024) draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve

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more problems given many samples (Yue et al., 2025; Wu et al., 2025; Oh et al., 2026). The two readouts agree for a model that samples from the distribution it states. Work outside medicine suggests that aligned models do not: they can verbalize distributions they fail to sample from (Meister et al., 2025; Zhang et al., 2026), and their samples can homogenize enough to defeat sampling-based uncertainty (Liu, 2026). Whether post-training opens this gap, and what it does to a differential diagnosis, where a reference distribution exists, has not been measured.

We measure it with a design that holds the pretrained weights fixed and varies only the post-training: base/instruct pairs for Qwen2.5-7B, Qwen3-8B and Qwen2.5-32B; every released stage (SFT, DPO, RL) of OLMo 3 7B (Team Olmo et al., 2025) and OLMo 3.1 32B, plus a 7B checkpoint trained with RL directly from the base; and three medical models fine-tuned from Qwen2.5-7B-Instruct. On DDXPlus (Fansi Tchango et al., 2022), which ships a reference differential for every synthetic patient, we score both readouts against that reference; on Phenopacket Store (Danis et al., 2025) we score rare-disease accuracy and stated confidence. Our findings:

- **Post-training collapses the sampled differential** at 7–32B: post-trained models gain at pass@1 but trail their base models at pass@32, give far fewer distinct answers, and drop can't-miss conditions from their samples. Which stage narrows most depends on the recipe: RL after SFT and DPO in OLMo 3 7B, SFT in OLMo 3.1 32B (§4.1).
- **The stated differential does not collapse; it improves.** One stated list moves closer to the reference and names severe conditions more often, while the samples move away, in every family and size; for OLMo 3 7B only when asked for a probability on every condition (§4.2).
- **Sampling-based methods lose their value; which signal to trust is model-dependent but cheap to check.** Majority voting stops helping, one list's top three beats 32 samples for post-trained Qwen models, and agreement weakens as an error signal. Whether the stated probability overtakes it is predicted by how often a model's samples all agree and how informative its stated probability is, and a 100-case audit picks the better signal in 88% of draws. Neither readout tracks a case's true ambiguity (§4.3).
- **Medical post-training adds accuracy, not breadth, and raises overconfidence on rare disease** (§4.4).

2. Related work

RL and the sampled distribution. RL with verifiable rewards improves pass@1 while base models win at large k on math and code (Yue et al., 2025), reweighting solutions already in the base model's support (Wu et al., 2025); a "sharpening tax" appears across 14 base/post-trained pairs on agentic tasks (Oh et al., 2026), though RL can expand the boundary in domains under-represented in pretraining (Cheng et al., 2025). In medicine, the one pass@k analysis we know of studies vision-language models (Jeddi et al., 2026). Instruction tuning and RLHF reduce output diversity (Kirk et al., 2024; West and Potts, 2025; Jiang et al., 2025); Zhang et al. (2026) attribute this to typicality bias in preference

data and track it across Tülu 3's SFT, DPO and RLVR checkpoints on creative writing. All of these measure the sampled distribution.

Stated versus sampled distributions. Language models asked to verbalize a group's opinion distribution match it better than their samples or log-probabilities do (Meister et al., 2025); that study uses instruct and API models on opinion surveys, which have no correct answer. Verbalized sampling recovers part of the diversity lost in post-training (Zhang et al., 2026), RLHF models' verbalized confidence is better calibrated than their token probabilities (Tian et al., 2023), and aligned models' samples often fall into one semantic cluster, leaving sampling-based uncertainty without signal (Liu, 2026). We add a task with a known reference distribution, controlled base/post-trained comparisons showing that post-training opens the gap, and its consequences for diagnostic aggregation and uncertainty.

Medical models and differential diagnosis. Medical post-training recipes include verifier-guided SFT and PPO (Chen et al., 2025), rule-based RL (Liu et al., 2025) and reasoning distillation (Huang et al., 2026); RL and reward models tend to favor confident answers (Leng et al., 2025; Damani et al., 2026; Mei et al., 2026). LLM evaluations on DDXPlus score hierarchical top-k accuracy (Lim et al., 2025), and Puri et al. (2026) train models to output several answers per completion, rewarding coverage of a gold set. We compare stated and sampled differentials against the reference *distribution*.

3. Setup

Tasks. **DDXPlus** (Fansi Tchango et al., 2022): synthetic patients from a rule-based medical knowledge base, each with a true pathology and a reference differential, a probability distribution over 49 conditions. We use 980 test patients (20 per condition) rendered as vignettes. The prompt lists the 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** (Danis et al., 2025): 1,401 published rare-disease cases covering 749 diseases, rendered as phenotype vignettes with an open-ended differential graded by mapping names to Mondo.

Models. Qwen2.5-7B, Qwen3-8B and Qwen2.5-32B, each with its official post-trained version; OLMo-3-1025-7B and OLMo-3-1125-32B with the SFT, DPO and final RLVR (Instruct) checkpoints of OLMo 3 7B and OLMo 3.1 32B, plus OLMo-3-7B-RL-Zero-General, trained with RL from the 7B base; and AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation), all fine-tuned from Qwen2.5-7B-Instruct, plus AlphaMed's checkpoint trained from Qwen2.5-7B (Table 4).

Sampling. 32 samples per case with vLLM, at T=0.7 (top-p 0.95) and, for the 7–8B models, T=1 (top-p 1), each model with its own chat template (OLMo base models get their SFT checkpoint's) and the same prompt. A sample is *valid* if its last JSON list names at least one condition; invalid samples count as wrong. They affect mainly base models (up to 47% of samples), so base-model coverage is a lower bound (Appendix B.3).

Two readouts. The *sampled* readout is the distribution of top-1 answers over a case's samples. We report pass@k (Chen et al., 2021), the number of distinct top-1 answers, the reference mass they cover (coverage@32), total-variation distance (TV) of their empirical distribution to the reference, and *Considered@32*: for conditions the reference weights at

least 0.1, whether the condition is the top answer of at least one sample, by DDXPlus severity. The *stated* readout is one sample's list, with its stated probabilities renormalized over the conditions it names (uniform if missing). We report its TV to the reference, the reference mass of the conditions it names, and whether it names each severe condition the reference weights at least 0.1, averaged over valid samples. Differences come with 95% paired case-bootstrap intervals.

4. Results

4.1. Post-training collapses the sampled differential

Table 1 compares each base model with its post-trained versions. In both Qwen families at 7–8B the post-trained model is far better on its first answer (+12.6 to +27.5 points), but its pass@k curve crosses the base model's at $k{\approx}4$–8, and at $k{=}32$ the base model leads by 9.7–14.6 points (all intervals exclude zero; Figure 2 in Appendix B.1). At 32B the pattern holds with smaller margins: Qwen2.5-32B-Instruct gains 10.4 points at $k{=}1$ and loses 4.9 $[2.8, 7.1]$ at $k{=}32$, and OLMo 3.1 32B Instruct loses 4.0 $[2.0, 5.9]$. Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3 for their base models, and cover less of the reference mass. Can't-miss conditions are not spared: severe-tier Considered@32 falls by 6.1–18.4 points, and conditions such as unstable angina and myocarditis almost vanish from the top answer (Appendix B.2). Which stage narrows the samples most depends on the recipe. In OLMo 3 7B, distinct answers fall from 4.93 to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR, and the RLVR step alone costs 19.1 $[16.3, 21.8]$ points of pass@32 against the DPO model. In OLMo 3.1 32B, SFT costs most (12.1 $[9.9, 14.4]$ points), DPO recovers most of it, and the RLVR step costs 2.8 $[1.1, 4.3]$ (Appendix B.5). RL-Zero, trained with RL from the 7B base, narrows less and keeps the base model's pass@32 at $T{=}1$ (-1.8 $[-4.5, +0.8]$; Appendix B.4). The pattern survives a reworded prompt and conditioning on valid samples; a chain-of-thought prompt removes the large-k loss but not the loss of diversity (Appendix B.3).

4.2. The stated differential does not collapse

For the post-trained models the two readouts move in opposite directions (Figure 1a, Table 2). For every Qwen model at 7–32B and at both temperatures, the post-trained model's single list is closer to the reference than the base model's (TV -0.043 to -0.086, all intervals exclude zero), covers more of its mass (+6.1 to +10.8 points), and names severe conditions more often (+9.0 to +13.4 points), while its sampled answers move farther from the reference (+0.029 to +0.071). OLMo 3.1 32B behaves the same way (Instruct: list TV -0.045, samples +0.021). Within each of these post-trained models the list beats its own samples by 0.04–0.09 in TV; within each base model the gap is about zero or favors the samples. Conditions that drop out of the top answer remain lower in the list: Qwen2.5-7B-Instruct names unstable angina as its top answer in 1 of 107 eligible cases but still lists it in half of them. OLMo 3 7B is the exception with the five-item prompt: its DPO and Instruct lists are no closer to the reference than its base model's (TV +0.000 and +0.028), and only their naming of severe conditions improves. Asked for a probability on every condition, its Instruct model's stated distribution is closer to the reference than the base model's (-0.032

Comparison	Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.322**	**0.34**
	post-trained	**0.558**	0.607	1.27	0.217	0.24
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**0.407**	**0.45**
	post-trained	**0.556**	0.643	1.53	0.236	0.27
Qwen2.5-7B, T=0.7	base	0.352	**0.695**	**4.91**	**0.335**	**0.40**
	instruct	**0.478**	0.598	1.82	0.239	0.28
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**0.422**	**0.51**
	instruct	**0.464**	0.663	2.65	0.290	0.34
Qwen2.5-32B, T=0.7	base	0.500	**0.795**	**3.78**	**0.352**	**0.35**
	instruct	**0.604**	0.746	1.68	0.276	0.29
OLMo 3 7B, T=0.7	base	0.178	**0.705**	**4.93**	**0.351**	**0.41**
	SFT	0.222	0.559	4.01	0.257	0.37
	SFT+DPO	**0.324**	0.613	3.30	0.272	0.34
	SFT+DPO+RLVR	0.266	0.422	2.37	0.195	0.25
	RL-Zero (from base)	0.295	0.572	4.06	0.286	0.33
OLMo 3.1 32B, T=0.7	base	0.326	**0.796**	**4.32**	**0.347**	**0.39**
	SFT	0.461	0.674	2.94	0.265	0.27
	SFT+DPO	**0.584**	0.784	2.29	0.293	0.32
	SFT+DPO+RLVR	0.560	0.756	2.15	0.283	0.30

Table 1: The sampled readout on DDXPlus (980 cases, 32 samples). Distinct: distinct top-1 answers per case; Cov@32: reference mass they cover; Severe C@32: share of severe conditions (reference probability ≥ 0.1) named as top answer in at least one sample. Bold: better model of each pair (OLMo: of the block). Invalid samples count as wrong, which mostly penalizes base models.

$[-0.038, -0.027]$), and the Qwen pairs reproduce their five-item results (TV -0.052 and -0.067; Appendix C.1). The stated-list result thus does not depend on the list format. The medical descendants' lists change little from their parent's (TV -0.014 to $+0.017$). In absolute terms every model is far from the reference: one list covers 0.22–0.42 of the reference mass, against 0.74 for the reference's own top five.

4.3. Consequences for sampling-based methods

Table 3 asks what the gap does to two uses of repeated samples. The single-call answer is the top answer of a model's first valid sample.

Aggregation. Majority voting over 32 samples (self-consistency) raises base models' accuracy by 3.4–8.4 points at T=0.7, but adds at most 1.9 for the general post-trained models (Qwen3-8B +0.3, Qwen2.5-7B-Instruct +0.8, OLMo-3-7B-Instruct -0.2, OLMo-3.1-32B-Instruct +0.9, all n.s.; Qwen2.5-32B-Instruct +1.9). Models that reason before answering still gain (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with a CoT prompt +4.5). Averaging stated lists instead of voting changes accuracy by at most about 2 points. For a top-three differential, the list is the better source after post-training for the Qwen models: the top three of one list contains the true pathology more often than the three most frequent of 32 sampled answers by 13.7 $[11.4, 15.9]$ points for Qwen3-8B, 7.3 $[5.3, 9.6]$ for Qwen2.5-7B-

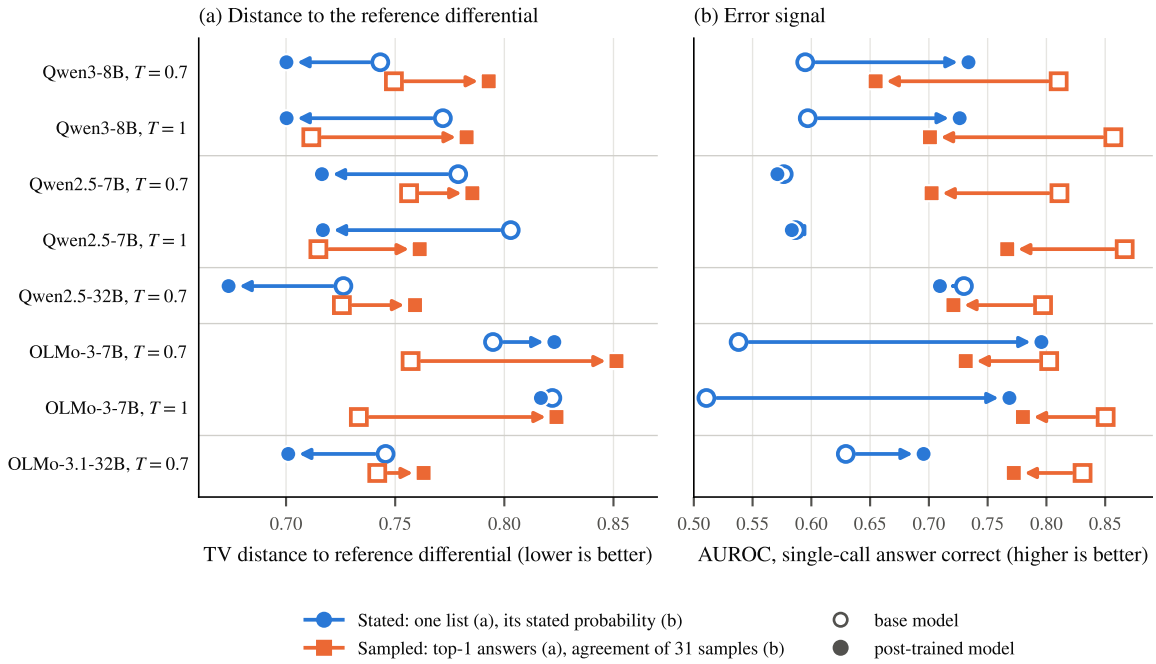


Figure 1: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; 32B models at T=0.7). (a) TV distance to the reference differential of one stated list (blue circles) and of the sampled top-1 answers (orange squares): post-training moves the list toward the reference and the samples away from it, except for OLMo 3 7B's five-item lists. (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange). Agreement weakens in every family; the stated probability overtakes it for Qwen3-8B and OLMo 3 7B at T=0.7.

Instruct and 5.6 [3.6, 7.7] for Qwen2.5-32B-Instruct, while for every base model the samples win by 7.3–13.6 points.

Uncertainty. We score each signal by how well it ranks the correctness of the single-call answer (AUROC). The stated probability needs no extra calls; *agreement*, the share of other samples with the same top answer, needs 4 or 31. For base models agreement is far better (0.80–0.83 against 0.54–0.73). Post-training weakens it in every family (Qwen3-8B 0.810 → 0.655, Qwen2.5-7B 0.811 → 0.702, OLMo 3 7B 0.802 → 0.731, Qwen2.5-32B 0.797 → 0.721, OLMo 3.1 32B 0.831 → 0.772; Figure 1b), consistent with Liu (2026). For Qwen3-8B and OLMo-3-7B-Instruct the stated probability is then the better signal (+0.079 [0.046, 0.110] and +0.064 [0.027, 0.101] over 32-sample agreement); for Qwen2.5-32B-Instruct it ties 32-sample agreement and beats five-sample agreement; for Qwen2.5-7B-Instruct, OLMo-3.1-32B-Instruct and the medical models, agreement stays better. AUROCs are on each model's own answers and compare signals within a model.

Comparison (T=0.7)	Model	TV to reference↓ One list	TV to reference↓ Samples	Reference mass covered One list	Reference mass covered 5 samples	Severe named One list
Qwen3-8B	base	0.743	**0.749**	0.335	**0.245**	0.347
	post-trained	**0.700**	0.793	**0.396**	0.207	**0.437**
Qwen2.5-7B	base	0.779	**0.756**	0.291	**0.233**	0.345
	instruct	**0.716**	0.785	**0.364**	0.214	**0.440**
Qwen2.5-32B	base	0.726	**0.726**	0.352	**0.272**	0.382
	instruct	**0.674**	0.759	**0.418**	0.245	**0.472**
OLMo 3 7B	base	0.795	**0.757**	0.268	**0.237**	0.256
	SFT+DPO	0.795	0.804	0.277	0.193	**0.378**
	SFT+DPO+RLVR	0.823	0.851	0.257	0.150	0.317
	RL-Zero (from base)	**0.775**	0.793	**0.295**	0.197	0.274
OLMo 3.1 32B	base	0.746	**0.742**	0.335	**0.254**	0.371
	SFT+DPO	**0.694**	0.755	**0.397**	0.249	**0.423**
	SFT+DPO+RLVR	0.701	0.763	0.391	0.241	0.392
Oracle	reference	0.265	0.18–0.25	0.735	0.563	0.975

Table 2: Stated vs. sampled readouts on DDXPlus (T=0.7, 980 cases, valid samples). *One list*: one sample's stated differential (mean over samples). *Samples*: empirical distribution of top-1 answers; *5 samples*: reference mass of the distinct top answers of five random samples, the same five-name budget as one list. Bold: better model of each pair (OLMo: of the block). Oracle: the reference's own top five as a list, and draws from the reference as samples. T=1 is in Figure 1.

Which signal to trust. Across 56 model–setting combinations (the models above at both temperatures and three prompts, our GRPO runs, and a full-distribution prompt), the stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all agree (Spearman 0.75), and with the stated probability's own AUROC (0.82); the stated probability wins for 19 of the 56. No single-answer threshold separates the two groups, but a small audit does: choosing the signal with the higher AUROC on 100 random cases with five samples each matches the choice made on all 980 cases in 88% of 500 draws per model, and the mean AUROC lost is 0.004, because the misses occur where the two signals nearly tie (Appendix C.2).

Ambiguity. The reference differential also measures how ambiguous a case is. For every model and signal, the Spearman correlation between uncertainty and the entropy of the case's reference differential lies between −0.11 and +0.16, except AlphaMed's stated probability (+0.30). By either readout, a model's uncertainty says almost nothing about a case's ambiguity.

4.4. Medical post-training and rare disease

Medical post-training on Qwen2.5-7B-Instruct adds accuracy without adding to the collapse. AlphaMed (rule-based RL) improves on its parent at every k (+3.9 at k=1, +8.4 at k=32) with unchanged sampled diversity (1.87 vs. 1.82 distinct answers); HuatuoGPT-o1 and m1, both trained on reasoning traces, widen the sampled differential (2.49 and 3.29) and raise severe-tier consideration by 8.4 and 13.5 points. AlphaMed's checkpoint trained from the

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
Qwen2.5-32B	0.510	+7.0	−8.3	0.730	0.759	**0.797**
Qwen2.5-32B-Instruct	0.598	+1.9	+5.6	0.709	0.663	**0.721**
OLMo-3 7B base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3 7B SFT+DPO	0.327	−0.4	−7.8	0.738	0.702	**0.751**
OLMo-3-7B-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
OLMo-3.1 32B base	0.441	+8.4	−13.2	0.629	0.805	**0.831**
OLMo-3.1-32B-Instruct	0.562	+0.9	−0.6	0.696	0.712	**0.772**
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 3: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Vote@32 gain: accuracy of the majority top-1 over all valid samples minus single-call accuracy (points). Top-3: rate at which one list's top three contains the true pathology minus the rate for the three most frequent sampled answers (points). AUROC: stated probability of the single-call answer (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal.

base model keeps most of the base model's diversity (3.14 distinct answers) and its can't-miss consideration (-0.7 $[-3.0, +1.7]$; Appendix D.1). On 1,401 Phenopacket Store cases every model is near the floor (pass@32 2.5–7.0%), but stated confidence is not. Medical RL raises it from 0.51 for the parent to 0.67–0.71 (ECE 0.49 → 0.64–0.70) while accuracy gains stay under 1.5 points, and every OLMo 3 7B stage states 0.73–0.84 at about 1% accuracy. Accuracy is also far higher on cases published before 2023 than after, consistent with memorization (Appendix D.2).

5. Discussion and limitations

Post-training does not simply narrow a model's differential: across two families and 7–32B, it improves the list the model states while collapsing the answers it samples. Evaluations that read the differential from samples (pass@k at large k, sample diversity, self-consistency, agreement-based uncertainty) therefore report a loss that a clinician reading one answer would not see, and they lose their value as tools on exactly the models deployed. The stated list is not a free replacement: it overtakes agreement as an error signal for only two of five general post-trained models, every model remains far from the reference differential, and neither readout tracks how ambiguous a case is. For a given model, though, a 500-call audit tells which signal to use. Which post-training stage narrows the samples most is a property of the recipe, not of SFT, DPO or RL as such: RL after SFT and DPO in OLMo 3

7B, SFT in OLMo 3.1 32B. Sampled diversity also still matters for training, since GRPO learns nothing from a group whose samples agree; in Appendix E we show that GRPO from Qwen3-8B-Base nearly matches the official model's top-1 accuracy while keeping more than twice its sampled diversity, and that a group-level reward narrows the samples less than standard RLVR.

Limitations. DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, so TV and coverage are distances to a *reference* posterior, not to clinical truth. Base models often give invalid answers (27–47% for OLMo), which we count as wrong; this makes large-k losses conservative but inflates small-k gains, and for OLMo no case has 32 valid base samples. The stated readout renormalizes each list over the conditions it names and uses the first valid sample as the single call; a full-distribution prompt gives the same picture, but other elicitations may differ. We study open 7–32B models from two families, one RL-Zero checkpoint (7B, trained with an LM-judge reward on chat prompts), and the 32B models only at T=0.7. Phenopacket cases come from published literature and may be in pretraining data. We do not evaluate clinical utility with clinicians.

Statement on AI Use. An AI assistant (Claude, Anthropic) was used to write and run analysis code, to draft and edit text, and to check citations. The authors designed the study, reviewed all code, results and text, and take responsibility for the content.

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Appendix A. Models

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B (Liu et al., 2025)	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B (Chen et al., 2025)	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K (Huang et al., 2026)	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B (Team Olmo et al., 2025)	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 4: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

Appendix B. The sampled differential in detail

B.1. General post-training trades coverage for top-1 accuracy

Table 5 and Figure 2 compare each base model with its post-trained counterpart on DDX-Plus. In both families the post-trained model is much better on its first answer: +27.5 points

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 5: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

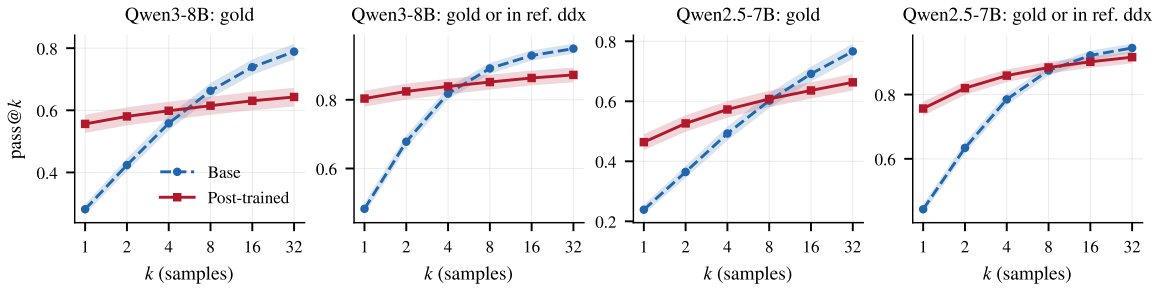


Figure 2: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.2 points at k=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.3, +12.0]$.

B.2. The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, post-trained models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.3 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.59 for base models and 1.87 for the reference differential (Table 5). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in

the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.2).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 6), and the severe "can't-miss" tier is not spared. For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, which is a defensible triage bias but not a differential.

B.3. Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 7). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains $+23.0$ $[+20.9, +25.1]$ points at k=1 and loses -7.1 $[-9.3, -5.0]$ at k=32 (T=0.7; at T=1, $+34.2$ and -8.7 $[-11.0, -6.4]$), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 ($+4.7$ $[+2.9, +6.6]$ at k=32) and ties it at T=1 (0.0 $[-2.1, +1.9]$). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§D.1).

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.32	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.32 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.32 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 6: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the can't-miss tier; medical post-training raises it; our GRPO from the base model loses at most 4 points, and with the group reward the severe-tier loss is not significant.

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, T=0.7), +18.7 [+16.6, +20.7] (Qwen3, T=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

B.4. Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 (Team Olmo et al., 2025) releases every stage of its 7B instruct pipeline from

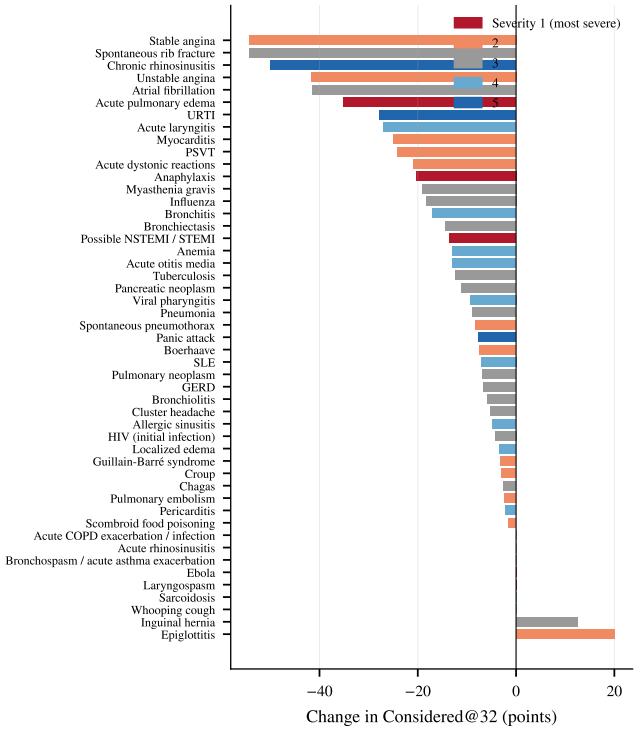


Figure 3: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 8 gives the results.

The coverage loss replicates in this third family. At T=0.7, every post-trained stage is worse than the base at k=32: by -14.6 $[-17.5, -11.7]$ points after SFT, -9.2 $[-12.1, -6.1]$ after DPO, -28.3 $[-31.5, -25.0]$ after RLVR, and -13.3 $[-16.1, -10.5]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k ($+5.4$ $[+2.9, +8.0]$ points at k=32, and $+10.3$ $[+9.0, +11.6]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-21.8, -16.3]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (-5.9 $[-7.2, -4.6]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-18.6, -13.5]$ points relative to the base,

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.559**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 7: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

against -7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: -12.7 $[-15.4, -9.9]$ at k=32).

RL applied directly to the base narrows the differential less. At T=1, RL-Zero keeps the base model's pass@32 (-1.8 $[-4.5, +0.8]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[1.2, 6.9]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32, about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §B.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (-1.9 to $+4.0$ points at T=0.7, $+1.3$ to $+8.1$ at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across three families, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§B.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero, AlphaMed's base-initialized checkpoint, and our GRPO runs in §E) narrows the differential less. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (-2.0 $[-3.0, -1.1]$ points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 $[-1.4, +0.2]$), from a base model at 4.5% (§D.2).

B.5. Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 9). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 $[2.8, 7.1]$ points at pass@32 for Qwen2.5-32B

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.266	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 8: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 6. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 9: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.2). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.3). Bold: best value per family.

and 4.0 $[2.0, 5.9]$ for OLMo 3.1 32B, halves the number of distinct answers (3.78 → 1.68 and 4.32 → 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 -12.1 $[-14.4, -9.9]$, distinct answers -1.39, severe tier -11.9); DPO recovers most of it ($+10.9$ against SFT), and the final RL step is small (-2.8 $[-4.3, -1.1]$ at k=32, -0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. The stated readout improves in both 32B families (§4.2), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (n.s.).

Model	TV to reference↓ One distribution	TV to reference↓ Samples	AUROC, single-call answer correct Stated p	AUROC, single-call answer correct Agree@5	AUROC, single-call answer correct Agree@32	Top-3: one dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 10: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

Appendix C. The stated readout: elicitation and audit

C.1. Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 10). The results match the short list. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038). Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) but not for the other two models.

C.2. Which readout to trust: a cheap audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell cheaply. We pool all 56 model–setting combinations we scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. Across them, the stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), and with the stated probability's own AUROC (0.82); it correlates negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. A small audit settles it for a given model. Using only five samples per case on 100 random cases, choose the signal with the higher AUROC. Across 500 draws per model, this choice matches the one made on all 980 cases in 88% of draws on average, and when it does not,

the two signals are close, so the mean AUROC lost is 0.004 (50 cases: 83%, 0.009; 200 cases: 92%, 0.002). The worst model is OLMo 3.1 32B DPO (43%), whose two signals tie (0.693 vs. 0.694).

Appendix D. Medical post-training and rare disease

D.1. Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): +3.9 points at k=1 ([+2.2, +5.5]), widening to +8.4 at k=32 ([+6.0, +10.6]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.29 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 6). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 11). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 [+11.3, +16.0] and 18.1 [+15.5, +20.7] points. m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper (Liu et al., 2025) describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 11, lower block). It does not. RL from the base model improves on the base at every k (+14.3 [+13.1, +15.7] points at k=1, +9.5 [+7.4, +11.6] at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at k=1 and 10.8 [+8.6, +13.2] points better at k=32. It also preserves can't-miss consideration: severe-tier Considered@32 changes by −0.7 [−3.0, +1.7] points relative to the base, against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 6). This is consistent with our own controlled GRPO-from-base result (§E): with a verifiable correctness reward, RL from a base model buys accuracy without the collapse, which is introduced by general instruction tuning and inherited by everything trained on top of it.

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 11: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best pass@32 of any 7B model.

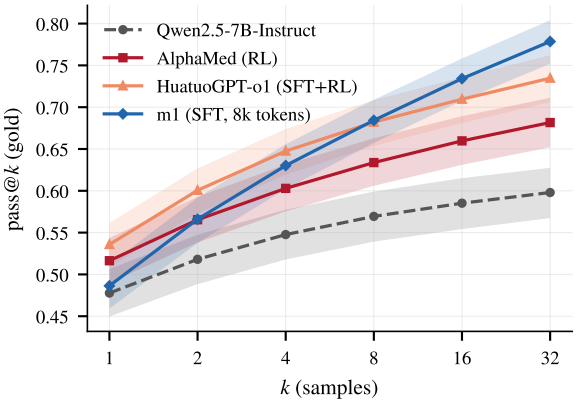


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: there is no sharpening tax at the medical stage.

OLMo 3 refines this (§B.4): within its instruct pipeline, the RL stage applied after SFT and DPO is the largest single narrowing.

D.2. Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 is 3.7–7.0%. Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67. Medical RL thus increases overconfidence exactly where the model is out of its depth. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident model we evaluate (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little

Model	≤2022 (n=901) pass@1	≤2022 (n=901) pass@32	2023–24 (n=348) pass@1	2023–24 (n=348) pass@32	2025+ (n=152) pass@1	2025+ (n=152) pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 12: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases, consistent with memorization of older case reports.

accuracy: AlphaMed beats its parent at every k (+0.6 [+0.2, +0.9] points at k=1, +1.0 [+0.1, +2.1] at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from k=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§E) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published, which suggests memorization (Table 12). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). Even the low absolute accuracy is therefore inflated by older, likely-memorized cases.

Appendix E. Training without collapsing the samples

The stated list survives post-training, but sampled diversity still matters: self-consistency and other sampling-based methods depend on it (§4.3), and so does reinforcement learning itself, since GRPO learns nothing from a group whose samples agree. Can reinforcement learning raise diagnostic accuracy without collapsing the sampled differential? We fine-tune

Qwen3-1.7B with GRPO on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

Setup. We train LoRA adapters (r=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with a common aggressive configuration (learning rate 2×10^{-5}, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use learning rate 5×10^{-6}, KL coefficient 0.04, no reward standardization (as in Puri et al. (2026)), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

Rewards. Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §B.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of n samples for a case. If sample i's top-1 answer is a, and k_a samples in the group give a,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\,p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where p_{ref} is the reference differential. This is sample i's marginal contribution to $n\sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution q and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is over-represented. In contrast to multi-answer rewards that score a set inside one completion (Puri et al., 2026), `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. As baselines we adapt the multi-answer rewards of Puri et al. (2026) to our prompt. Multi-RLVR scores the share of listed conditions that belong to the reference differential, treated as a set of acceptable answers; Multi-RLCR adds their set-level Brier term on the per-answer probabilities. Both zero the reward for duplicate or invalid answers, as in the original, and use the same format reward, optimizer and settings as our recipes.

Results. Table 14 reports means over two seeds. All recipes raise top-1 accuracy by 8–10 points. The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §B.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by +1.6 [+1.0, +2.3] and pass@32 by +1.7 [+1.0, +2.6] points;

- distinct answers by $+0.10$ $[+0.07, +0.13]$ and entropy by $+0.034$ $[+0.022, +0.047]$;
- coverage@32 by $+1.5$ $[+1.0, +2.1]$ points, and TV by -0.014 $[-0.019, -0.010]$.

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 $+1.0$ $[+0.6, +1.6]$, entropy $+0.013$ $[+0.005, +0.021]$). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The set-membership baselines of Puri et al. (2026) train stably (their reward rises from 0.27–0.31 to 0.32–0.35) but barely change sampled behavior. pass@1 rises by under 1 point over the untrained model, and diversity is unchanged. Against them, $\texttt{gm}^+$ is better on every metric, by $+8.9$ $[+7.4, +10.5]$ points at pass@1, $+0.07$ $[+0.03, +0.11]$ distinct answers and $+3.0$ $[+2.3, +3.8]$ points of coverage@32 (vs. Multi-RLVR; Multi-RLCR is similar). What they do improve is the differential written *inside* one completion: the reference mass covered by a single listed differential rises to 0.317 (Multi-RLVR), against 0.300 for `acc` and 0.309 for $\texttt{gm}^+$. The two objectives are complementary: a set reward shapes the list a model writes, while the group reward shapes which answer it commits to across samples, which is what collapses under post-training. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor; the same holds at 8B (§D.2).

Starting from a diverse base model. The 1.7B experiment starts from an already collapsed instruct model. We therefore repeat the comparison from Qwen3-8B-Base, the diverse base model of §B.1, with the same recipe and `acc` vs. `gm`. We compare against Qwen's own post-trained Qwen3-8B (Table 13; means of two seeds). Both RL recipes reach nearly the official model's top-1 accuracy (0.537 and 0.539 vs. 0.558; `gm`: -1.9 $[-3.9, +0.1]$ points). They keep more than twice its diversity (2.7–2.9 vs. 1.27 distinct answers) and beat it by 15–16 points at pass@32 (`gm`: $+16.3$ $[+14.0, +18.8]$). The collapse produced by general post-training is therefore not the price of its top-1 accuracy. Both runs still narrow the base model somewhat, and the group reward narrows it less. At equal pass@1 ($+0.2$, n.s.), `gm` beats `acc` on distinct answers $+0.14$ $[+0.11, +0.17]$, entropy $+0.035$ $[+0.029, +0.041]$, coverage@32 $+1.1$ $[+0.8, +1.5]$ points, and pass@32 $+1.2$ $[+0.5, +1.9]$. Each seed reproduces the ordering. The severe tier tells the same story (Table 6): official post-training removes 9.8 points of can't-miss consideration, `acc` 2.8, and `gm` 1.5 (not significant). Figure 5 shows the pass@k curves.

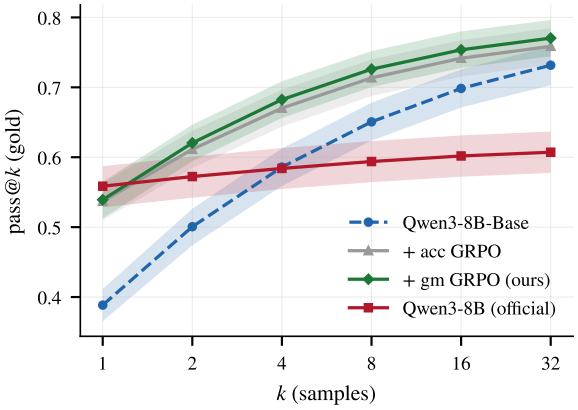


Figure 5: pass@k on DDXPlus (T=0.7, 980 cases) from the same pretrained weights: Qwen3-8B-Base, our GRPO runs (pooled over two seeds), and the official post-trained Qwen3-8B, with 95% case-bootstrap bands. Our RL lifts the whole curve; official post-training trades the right end for the left.

Model	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-8B-Base (start)	0.389	0.732	**3.77**	**0.78**	**0.323**	0.749
+ `acc` GRPO (ours)	0.537	0.759	2.72	0.52	0.297	0.750
+ `gm` GRPO (ours)	0.539	**0.770**	2.86	0.56	0.308	**0.742**
Qwen3-8B (official post-training)	**0.558**	0.607	1.27	0.11	0.217	0.793

Table 13: GRPO from Qwen3-8B-Base vs. the official post-trained Qwen3-8B on DDXPlus (980 cases, 32 samples, T=0.7); GRPO rows are means of two seeds. Our RL nearly matches the official model's top-1 accuracy while keeping more than twice its diversity; the group reward preserves more than `acc`.

Recipe	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-1.7B (untrained)	0.378	0.435	1.41	0.151	0.162	0.850
`acc`	0.458	0.499	1.36	0.142	0.181	0.829
`dist`	0.458	0.502	1.34	0.131	0.181	0.829
Multi-RLVR (Puri et al., 2026)	0.385	0.443	1.39	0.153	0.166	0.845
Multi-RLCR (Puri et al., 2026)	0.383	0.441	1.39	0.150	0.165	0.847
`gm`	0.469	0.510	1.40	0.155	0.186	0.825
`gm`$^+$	**0.474**	**0.516**	**1.46**	**0.176**	**0.196**	**0.815**

Table 14: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, T=0.7); means of two seeds. `gm`$^+$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; set-membership rewards barely move it; the group reward preserves, then widens, its differential while improving accuracy.